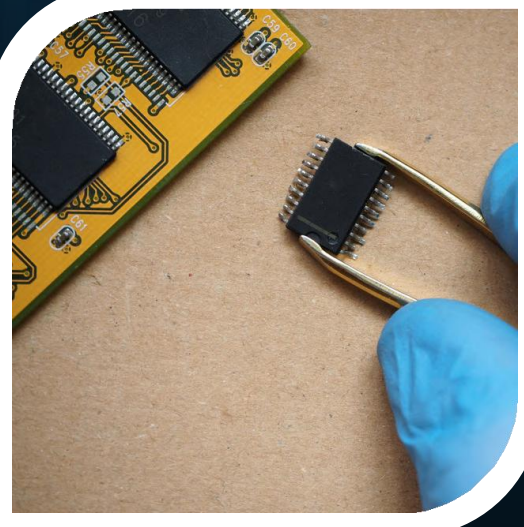
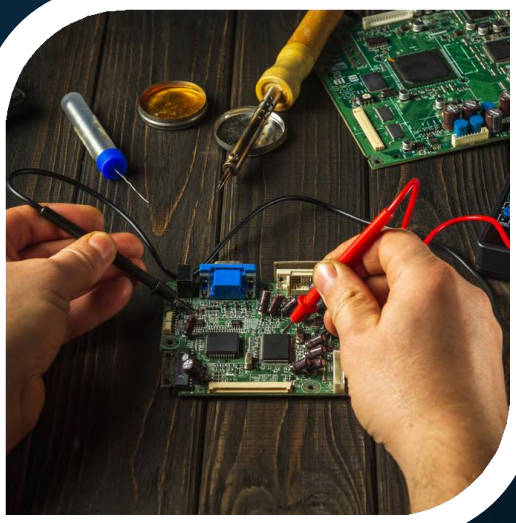
IIT Kanpur

POST GRADUATE PROGRAMME

# DEPARTMENT OF ELECTRICAL ENGINEERING

INDIAN INSTITUTE OF TECHNOLOGY KANPUR








www.iitk.ac.in/ee

# Electrical Engineering

The Department of Electrical Engineering at IIT Kanpur offers M.Tech., MS(R), and Ph.D. programmes spanning the breadth of modern Electrical Engineering. The Department has a strong interdisciplinary research culture, with activities ranging from fundamental theory and devices to intelligent systems, emerging technologies, and large-scale engineering applications.

Microelectronics and VLSI research encompasses semiconductor devices, nanoelectronics, analog, digital, RF and mixed-signal integrated circuits, VLSI system design, device modelling and simulation, neuromorphic computing, hardware security, non-volatile memories, spintronics, two-dimensional materials, organic and flexible electronics, and emerging semiconductor technologies.

Power Engineering covers power system dynamics, stability, operation, control, protection and state estimation; smart grids, electricity markets and renewable-energy integration; high-voltage engineering, dielectrics and insulation; and power electronics, electric machines and drives. Research also includes electric vehicles, wired and wireless EV chargers, wide-bandgap SiC/GaN power converters, EMI/EMC-compliant converters, wireless power transfer, renewable-energy interfaces and microgrids.

RF and Microwaves research includes antennas and phased arrays, metamaterials and metasurfaces, reconfigurable intelligent surfaces, RF and microwave circuits, mm-wave and THz technologies, microwave and tomographic imaging, RF sensors, electromagnetic compatibility, non-destructive testing, and RF energy harvesting and wireless power transfer.

Signal Processing, Communications and Networks spans information and coding theory, wireless and communication networks, 5G/6G technologies, massive MIMO, mm-wave and THz communication, signal and image processing, speech and multimedia processing, computer vision, inverse problems and tomography, optimization, machine learning, and networked information systems. These themes build on the communication and signal-processing areas already represented in the brochure.

Control and Automation covers control and dynamical systems, nonlinear, adaptive and robust control, networked and distributed control, multi-agent and cooperative systems, robotics and autonomous systems, guidance and control of unmanned vehicles, electric-vehicle control, and power-system control. Research increasingly extends control theory to emerging interdisciplinary domains such as microrobotics, cyber-physical systems, systems and synthetic biology, intelligent control, and learning-based control. The Department has dedicated experimental facilities for robotics, networked control, UAVs, electric vehicles, biological systems and microrobotics.

Optoelectronics and Optical Communication research includes integrated photonics, nanophotonics, plasmonics, quantum optics and quantum technologies, optoelectronic devices, optical communication and networks, nonlinear and specialty fibres, fibre-optic sensors, biomedical optics, optical imaging and metrology, and photonic devices and systems. These activities span both fundamental photonics and emerging applications in communication, sensing, imaging and quantum technologies.





# POST-GRADUATE PROGRAMMES OFFERED

In the application form for M.Tech., the applicants must specify their choice of area(s) of specialization/code number mentioned above. Please note that the candidates must use only those code numbers given at the end of this section while filling up application form for Ph.D. programme. Eligibility for a specialization may depend on the candidate's choice of test paper in the GATE examination. For detailed information regarding eligibility and minimum qualifications, applicants should refer to the website of the Dean of Academic Affairs (www.iitk.ac.in/doaa/). Candidates are also advised to check http://www.iitk.ac.in/ee/admissions for regular updates on EE admissions.

## M.Tech.

In M.Tech. programme, a student takes eight courses, some of which may be compulsory for the area of specialization chosen, the rest being electives to be chosen in consultation with the programme advisors. The programme also includes a research/project component, which culminates in a thesis.

## MS(R)

In MS(R) programme, a student has to complete a minimum of four courses in the area chosen by the student in consultation with the thesis advisor. After completing the course work, the student is expected to do research work leading to a thesis. Focus on research makes this program exciting and provides in-depth knowledge to the student.

## Ph.D.

In the Ph.D. programme, a student with M.Tech qualification has to complete minimum of four courses and a student with B.Tech. qualification has to complete minimum of ten courses. The most important part of the doctoral programme is the research work leading to a thesis. Student works on exciting research problems to come up with innovative/original ideas.

Specialization in the M.Tech/MS(R)/Ph.D. Programmes is available in any of the following broad areas:

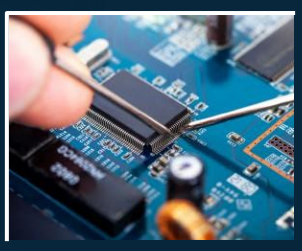
- Microelectronics and VLSI (Code: 01)
- Power Engineering (Code: 02)
- RF and Microwaves (Code: 03)
- Signal Processing, Communications and Networks (Code: 04)
- Control and Automation (Code: 05)
- Optoelectronics and Optical Communication (Code: 06)











# LABS/FACILITIES

The department has excellent research laboratories and support facilities in several areas.

Micro fabrication lab with basic semiconductor processing capability for silicon as well as organic material based devices (OLED, organic solar cells, OTFT, etc.); Solar cell characterization lab; photo mask making facility; Semiconductor device lab with capability to synthesize organic materials for organic LEDs and solar cells; Integrated circuits simulation and VLSI design laboratory with all the modern EDA tools, (e.g. Cadence, Synopsis, Mentor Graphics, Xilinx based gate array design & programming tools, etc.) and adequate hardware in the form of servers and good number of workstations for research and course work with provision to fabricate chips at different technology nodes.

Three teaching/training labs have been developed to train students in areas related to organic electronics. These are the organic electronics processing lab, the organic electronics characterization lab and the organic electronics simulation lab. Robotics lab equipped with 7 DoF manipulators, mobile robots, and visual systems for autonomous navigation of mobile robots, multi-robot formation and control.

Control system lab with facilities for microprocessor-based control of PMDC motors, multimotor coordination, networked control and control of electric vehicles. Distributed Systems and Control (DiSCo) Lab is equipped with quadcopters, fixed wing UAVs, flight simulator, processor in-loop testing facility for quadcopters, simulation of multi-agent systems.



Quantum Key Distribution Lab

The Dynamics and Control Lab focuses on the multidisciplinary aspects of control theory and engineering, primarily in systems and synthetic biology, microrobotics, and power system control. The lab is equipped with state-of-the-art molecular biology equipment to design genetic constructs and characterize their performance both in vivo and in vitro. It also features micro/nano robots, along with a magnetic manipulation system, simulation and computational platforms to carry out control engineering analysis and design.



Advanced Fiber-Optics Lab

The High Voltage Laboratory has AC, DC, impulse test facilities, partial discharge monitoring, time domain spectroscopy, frequency domain spectroscopy, PEA for space charge measurements, and an outdoor insulation test bay; Power electronics and static control laboratory with solid state control of electric drives; Power systems simulation laboratory equipped with Six Racks Real Time Digital Simulation (RTDS), Opal-RT and other modern simulation tools; NAMPET laboratory with complete fabrication and testing facilities for research in power electronics including frequency response analyzer, solar photovoltaic panels. Power management lab with solar simulator, frequency response analyzer, electronic loads and fabrication facility.



Optical Metrology and Imaging Lab

MEIPES, EmPower and Power Management laboratories are also equipped with Power Electronics prototype fabrication and testing facilities.

RF and Microwaves lab having network analyzers up to 110 GHz, spectrum analyzers, signal generators, power meters, noise figure meter, shielded anechoic chamber for antenna and RCS measurements, microwave imaging and material testing facility over a wide frequency range, dielectric probe kit, rectangular waveguide and coax calibration kits for various frequency bands.



Optoelectronics & Nanofabrication Lab

# LABS/FACILITIES

Fiber optics laboratory equipped with optical spectrum analyzer (600 nm-2000 nm) and interface development facility for fiber optic links, clean room for semiconductor optoelectronic device fabrication and photonic measurement laboratory.

In addition, Advanced Fiber optics laboratory has WDM optical components, semiconductor optical amplifiers, single-mode standard and nonlinear fibers, Sampling oscilloscope (optical 40 GHz and electrical 65 GHz bandwidth), electronics to enable experiments on 40-100G optical links.

Networks laboratory with scalable and configurable test-bed for simulating complex network topologies, 802.11 WiFi links, software radio, multiservice network and QoS, etc. Wireless Communications laboratory is equipped with NI USRPs platform. Quantum Photonics Research Laboratory is equipped with optical characterization facility, comprising of a home-built microscope attached to a high-resolution spectrometer and a CCD camera, for testing nanophotonic and plasmonic devices, and FDTD simulation tools for designing these devices.

Speech processing and multi-modal information processing lab equipped with the state of art multi-channel audio visual data acquisition test bed along with dedicated data and voice server connected on E1 digital telephony line enabling research on multi-channel and multi-modal information processing and content delivery;

Digital signal processing laboratory with multiple PCs and DSP hardware based on Texas instrument's DSPs; Communications laboratory equipped with USRPs and WARP boards. Computer vision lab equipped with chroma keying, controlled illumination, structured light sources, various kinds of camera and associated computational resources.



Tomographic Imaging Lab

Electronic equipment maintenance and calibration facility; Multilayer (up to six layers) PTH printed circuits fabrication facility, including CAD facility for printed circuits design and verification; Department library with a good collection of specialized books, research reports and data catalogues; An extensive campus wide LAN with a high speed internet connectivity.



Communication Networks Lab

5G testbed Lab is developed to build the testbed that closely resembles a real-world 5G deployment. The testbed will deliver an end-to-end 5G testbed comprising of 5G BS and UE nodes that support enhanced mobile broadband (eMBB), Ultra low latency communication (URLLC) and massive MTC including NBIoT services. This testbed could become a basis for many commercial deployments.



Anechoic Chamber

The wide-ranging research facilities and various sponsored research activities ensure that the students are thoroughly exposed to modern trends in Electrical Engineering. The informal atmosphere and free discussions between the students and the teachers are a source of inspiration to both the sides and maintain the standards of academic progress.



5G TESTBED LAB



EMI/ EMC Test Facility



RF Dielectric Testing Facility

# FACULTY LIST

- Akhtar M J, Ph.D. (University of Magdeburg): Microwave, mm-wave and THz imaging and non-destructive testing, Metamaterial inspired RF sensors including nano-biosensors, RF energy harvesting and wireless power transfer, FSS and nanomaterials based wideband microwave absorbers for stealth technology and EMI/EMC applications, RF amplifiers, High frequency plasmonic devices and applications, Metasurface and RIS based RF structures and systems.
- Arora V, Ph.D. (Indian Institute of Technology Kanpur): Audio signal processing, machine learning, automatic speech recognition, music information retrieval.
- Banerjee A, Ph.D. (University of Notre Dame): Error Control Coding, Wireless Communications, Applications of Machine Learning in communications and signal processing, and Molecular Communications
- Battula, Swathi, Ph.D. (Iowa State University): electricity markets, modelling and design of electrical energy systems, energy policy and management.
- Behera L, Ph.D. (Indian Institute of Technology Delhi): Intelligent control; Soft computing; Quantum computing and Information; Applied nonlinear control.
- Bhagat N A, Ph.D. (University of Houston): Neural & Bio-signal processing, Medical Instrumentation, Brain-machine interfaces, Functional Electrical Stimulation, and Rehabilitation Engineering.
- Bose A, Ph.D. (Indian Institute of Technology Bombay): Spin-based electronics, Fabrication and characterization of nanoscale devices, 2D material heterostructure, topological quantum materials, generation of unconventional spin current, altermagnets, MTJs, spincaloritronics, spin orbitronics, spin- mechatronics, chiral spin textures among others.
- Budhiraja R Ph.D. (Indian Institute of Technology Madras): Applications of linear algebra, information theory and optimization to study problems in wireless communications, massive MIMO, cooperative communications, 5G algorithm design, building communication systems.
- Chakrabarti S, Ph.D. (Memorial University of Newfoundland): Power system dynamics and stability; Power system state estimation; Synchrophasor applications in power systems; Power system reliability.
- Chaturvedi A K, Ph.D. (Indian Institute of Technology Kanpur): Communication theory and systems; Mobile communications; Spread spectrum systems.
- Chaudhary R K, Ph.D. (Indian Institute of Technology Kanpur): Reconfigurable Metamaterial/ Metasurface Antenna, Multifunctional MIMO-Cognitive-Radio Antenna, Phased Antenna Array, Dielectric Resonator Antenna, mmWave Circuits, Frequency Selective Surfaces and Intelligent Reflecting Surface.
- Chauhan Y S, Ph.D. (Ecole Polytechnique Federale de Lausanne): Nanoelectronics; Compact modeling of semiconductor devices; Low and high frequency electrical characterization; Atomistic Simulation; RF Circuit Design.
- Chithra, Ph.D. (Indian Institute of Technology Madras): Analog and mixed-signal VLSI design, digital VLSI design, frequency and phase synthesis, time-to-digital converters.
- Dey R, Ph.D. (University of Texas): Topological Insulators, Transition Metal Chalcogenides and Di-Chalcogenides, Fabrication, Characterization and Analysis of micro and nano devices.
- Dutta A, Ph.D. (Louisiana State University): Semiconductor device modeling; IC fabrication tehnology, Analog/digital/mixed-signal VLSI circuits.
- Ganeriwala Mohit, (Indian Institute of Technology Gandhinagar): Multiscale-Atomistic, Numerical and Compact Modeling, TCAD, Electrical characterization, Neuromorphic Computing, Two-dimensional Materials and Advanced nanoscale FETs.
- Gupta A K, Ph. D. (University of Texas): Wireless Communications, Stochastic Geometry and Numerical Methods.

# FACULTY LIST

- Gupta N, Ph.D. (Indian Institute of Science): High voltage engineering: Dielectrics and electric insulation; Nanodielectrics; Numerical techniques in electric and magnetic field computation.
- Gupta S, Ph.D. (University of Maryland College Park): Nanophotonics; Plasmonics; Quantum optics; Quantum dot based Devices.
- Gururaj M V, Ph.D. (Indian Institute of Technology Ropar): Renewable penetration challenges to the grid, Machine learning applications to power systems, EV interfacing challenges.
- Harish A R, Ph.D. (Indian Institute of Technology Kanpur): Antennas; RF Engineering; Electromagnetics.
- Hegde R M, Ph.D. (Indian Institute of Technology Madras): Multimedia information processing; Speech signal processing; Array processing; Application of signal processing in wireless networks.
- Iyer S S K, Ph.D. (University of California): Organic solar cell; semiconductor devices.
- Jagannatham A K, Ph.D. (UC San Diego): 5G/ 6G Wireless Technologies, Massive MIMO, mmWave MIMO, Non-Orthogonal Multiple Access (NOMA), OTFS (Orthogonal Time Frequency Space) Modulation, THz, IRS, VLC.
- Jerripothula K R, Ph.D. (NTU Singapore): Computer Vision, Artificial Intelligence & Machine Learning, Multimedia Signal Processing, Image Processing, and Healthcare Informatics.
- Kant P, Ph.D. (Indian Institute of Technology Delhi): Multi-winding transformers for multi-pulse AC-DC converters, Multi-level inverters for medium voltage/power applications, Modulation techniques for voltage source inverters, Medium voltage drives, Power electronics, Electrical machines and drives, Electric vehicle drive system.
- Lahgere A, Ph.D (Indian Institute of Technology Delhi): Semiconductor Devices, Compact Modeling, Emerging CMOS Devices, Emerging Volatile-Nonvolatile Memory, Variation, PPA Benchmarking,Analog/RF, and Neuromorphic computing.
- R.S. Ashwin Kumar, Ph.D. (Indian Institute of Technology Madras): Analog & mixed-signal integrated circuits and signal processing.
- Mazhari B, Ph.D. (University of Illinois at Urbana Champaign): Semiconductor device modeling and fabrication; VLSI design; Transducers and sensors.
- Mathew E C, Ph.D. (Indian Institute of Technology Delhi): HVDC and MVDC transmission systems, Fault tolerant modular converters for high-power applications, Medium and high-power DC-DC converters, Application of Power Electronics in Power Systems, Planning and operation of bulk power systems, Power Electronics dominated Grid.
- Mondal I, Ph.D. (Indian Institute of Technology Madras): Analog and RF integrated circuits.
- Mondal W U, Ph.D. (Indian Institute of Technology Kharagpur): Reinforcement Learning, Game Theory, Network Economics, Wireless Communication.
- Mohapatra A, Ph.D. (Indian Institute of Technology Delhi): Optimization for operation and planning of power networks; AC and DC microgrid protection; Stability, security and control of power networks with renewable resources.
- Naik N, Ph.D. (Indian Institute of Science): Reconstruction and analysis approaches to tomographic problems; Numerical solutions for wave propagation; Sub-surface imaging.
- Nallam N, Ph.D. (Indian Institute of Technology Delhi): Analog, RF, and mm-Wave Integrated Circuits
- Patel A, Ph.D. (Indian Institute of Technology Delhi): Control and Dynamical Systems, Systems and Synthetic Biology, Microrobots, Robust Control Theory, and Wide-Area Control of Power Systems.
- Potluri R, Ph.D. (University of Kentucky): Control system theory; Practical applications of control theory; Electric vehicles; Networked control systems; Consensus and Cooperation.
- Poluri N, Ph.D. (University of Sheffield): Design of broadband power amplifiers and front- end modules, Power amplifier linearization techniques, and phased array systems.

# FACULTY LIST

- Peruru S S, Ph.D. (Indian Institute of Technology Madras): Wireless Networks, Artificial Intelligence and Probabilistic Graphical Models.
- Kant P, Ph.D. (Indian Institute of Technology Delhi): Multi-winding transformers for multi-pulse AC-DC converters, Multi-level inverters, Modulation techniques, Medium voltage drives, Electrical machines and drives, Motor control algorithms for electric vehicles.
- Kundu U, Ph.D. (Indian Institute of Technology Kanpur): Applications of resonant converters, Power management ICs, Wireless power transfer technology.
- Pradeep Kumar K, Ph.D. (Indian Institute of Technology Madras): Signal processing for Optical Communications, Non linear fiber optics; Quantum cryptography; Fiber-optic sensors, Spin Waves.
- Rajawat K, Ph.D. (University of Minnesota): Optimization Algorithms, Network Optimization, Signal Processing in Networks, Online Machine Learning.
- Rajshekhar G, Ph.D. (Swiss Federal Institute of Technology Lausanne): Biomedical Optics; Light Microscopy; Optical Metrology; Digital Holography.
- Rituraj, Ph.D. (Stanford University): Integrated photonic circuits, Nanophotonics, Quantum optics and Optoelectronic devices.
- Sahay S, Ph.D. (Indian Institute of Technology Delhi): Hardware Platforms for Neuromorphic Computing, Hardware Security Primitives, Novel Device Architectures for Scaling CMOS Technology, Analytical and Compact Modeling of Semiconductor Devices, Nonvolatile Memories, and Spintronics.
- Samanta S., Ph.D. (Concordia University): Power Electronics, Electric Vehicles, Wireless Power Transfer, Resonant Converters with WBG devices.
- Sahoo S R, Ph.D. (Indian Institute of Technology Bombay): Linear and Nonlinear systems, Adaptive Control, Timedelay systems, Control of unmanned vehicles, Multi-agent systems and coordinated control,AC and DC Microgrids.
- Sandhan T, Ph.D. (Seoul National University): Computer vision, Machine learning and Robotics
- Sensarma P, Ph.D (Indian Institute of Science): Power electronic converters; Power quality; FACTS devices; Renewable energy delivery systems; Motor drives.
- Sharma A, Ph.D. (Indian Institute of Technology Kanpur): Power Systems, Smart Grid Technology, State Estimation, IT Application into Power Systems, Smart City, Multi-Agent Systems, Wide Area Monitoring & Control of Power System, Energy Market, Demand Response Management, Internet of Things.
- Singh S N, Ph.D. (Indian Institute of Technology Kanpur): Power system restructuring; FACTS technology; Optimal power dispatch and security analysis; Power system dynamics, operation and control; Power quality; Application of genetic algorithms and artificial neural networks in power systems; Wind power.
- Singh Y N, Ph.D. (Indian Institute of Technology Delhi): Telecommunication networks; Optical communications; Optical networks; Digital switching systems; Wireless networks; Wireless sensor networks; P2P networks.
- Srivastava K V, Ph.D. (Indian Institute of Technology Kanpur): Metamaterials, Microwave Antennas, Printed Antennas, MIMO Antennas, Microwave Metamaterial Absorbers, Microwave Cloaking, Finite Difference Time- Domain (FD-TD) Technique.
- Tripathy T, Ph.D. (Indian Institute of Technology Bombay): Guidance of autonomous vehicles, Formation control problems in multi- agent systems, Study of nonholonomic systems, Missile guidance.
- Vasudevan K, Ph.D. (Indian Institute of Technology Madras): Communication systems; Signal processing for communications.

# FACULTY LIST

- **Verma A, Ph.D. (University of Notre Dame): Epitaxial growth and characterization of semiconductor thin films, Semiconductor Device Design and Fabrication, Electron Transport measurements and modelling.**
- **Verma A R, Ph.D. (Indian Institute of Science): High Voltage Engineering, Dielectrics and Electrical Insulation, Computational Electrostatics, Polymeric Insulation for High Voltage Application, Over-voltages in Power Systems.**
- **Verma N K, Ph.D. (Indian Institute of Technology Delhi): Big Data; Internet of Things/Cyber physical systems; Intelligent Data Mining Algorithms and Applications; Diagnosis and Prognosis of Rotating Machines; Soft- Computing in Modelling and Control; Machine Learning Algorithms; Computer Vision; Bioinformatics; Smart Grid; Intelligent Agents and their Applications; Intelligent Informatics;** Fuzzy Controllers; Image frame generation; Brain Computer/Machine Interface.













# BROAD RESEARCH AREAS

- Digital Communication Systems; Information and Coding Theory; Telecom and Wireless Networks; Peer-to-peer networks; Digital Switching Systems; 5G Wireless Communication Systems; Digital Signal and Image Processing; Computer Vision; Inverse Problems and Tomography; Signals and Systems Theory.

- Control and Dynamical Systems and Robotics; Networked Control and Electric Vehicle Control; Electronic and Virtual Instrumentation; Fuzzy Logic; Neural Networks and their applications.

- Power Systems Economics; Optimization and Markets; Power System Protection; HVDC & FACTS, Power Quality; Smart Grid and Synchrophasors; Power Electronics; Electric Drives; Active Power Filters and Static VAR Systems; Nanodielectrics and thin film capacitors; Renewable Energy Interfaces.

- Microelectronics; VLSI System Design; Wired and Wireless EV Chargers; SiC/GaN-based Power Converters; EMI/EMC-complaint Power Converters. Analog and Digital Circuit Design; Semiconductor Device Modeling and Simulation; Solid State Devices; Nanoelectronics and Nano-scale Devices; Organic Electronics; Flexible Electronics; Photovoltaics Neuromorphic Computing, Hardware security, biomedical engineering and instrumentation, CMOS ICs for Software Defined Radios, Phased Arrays, Satcom, Radars, Data Converters, Synthesizers, spintronics, etc.

- Electromagnetics; RF Engineering and Microwaves; Antennas, Metamaterials; MMIC; RF and Microwave Sensors; RFID; Microwave and Mm-wave Imaging; RF Energy Harvesting, Electromagnetic and Tomographic Imaging; Terahertz Imaging and Testing.

- Nanophotonics, Plasmonics, Quantum Dot based Devices; Optoelectronics; Signal processing for fiber-optics; Nonlinear fiber optics; Fiber-optic sensors; Quantum Cryptography and Quantum Optics; Spin waves; Photonic Networks and Systems.









# CONTACT


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